

ML Problem Solving

Perceptron Learning Algorithm

In this problem-solving activity, teams/squads will compete against each other for ML dominance! Make sure you go over the write-up thoroughly and ask instructor for clarification if needed. This problem-solving has strict timing, so please make sure at least one team member keep an eye on the clock/timer!

Tournament Overview and Rules of Engagement

Welcome to the **ML Arena**. In this live competitive in-class activity, student squads compete across progressive engineering rounds.

The Master Tournament Scoreboard

Squads earn **Arena Points (AP)** across every module through direct problem solving, defense resilience, and adversarial attacks:

ARENA POINT (AP) DISTRIBUTION	
Round 1: Heuristic Defense (+100 AP)	Red-Team Heuristic Break (+150 AP)
Round 2: Ledger Precision (+150 AP)	Permutation Sabotage Bounty (+150 AP)
SPEED BONUS: First squad to finish a round with instructor sign-off (+50 AP)	

Competition Scenario: The Silicon Wafer Sensor Grid

Your squads are rival engineering firms competing for an industrial contract to build real-time defect detectors for semiconductor manufacturing. A high-speed 2×2 photodiode sensor array captures light transmitted through silicon wafers. Each pixel records an 8-bit grayscale intensity value $b_{r,c} \in \{0, 1, \dots, 255\}$, organized row-wise into an input vector:

$$\mathbf{x}^k = \begin{bmatrix} b_{0,0} \\ b_{0,1} \\ b_{1,0} \\ b_{1,1} \end{bmatrix} \in \mathbb{R}^4$$

The target label $a^k \in \{0, 1\}$ represents quality status ($a^k = 1$: Nominal / Passing; $a^k = 0$: Defective / Rejected).

The baseline calibration dataset $\mathcal{D} = \{(\mathbf{x}^k, a^k)\}_{k=1}^6$ is:

- $\mathbf{x}^1 = [255, 128, 128, 0]^\top, \quad a^1 = 0$
- $\mathbf{x}^2 = [55, 128, 128, 128]^\top, \quad a^2 = 0$
- $\mathbf{x}^3 = [192, 128, 128, 0]^\top, \quad a^3 = 0$
- $\mathbf{x}^4 = [100, 128, 128, 100]^\top, \quad a^4 = 1$
- $\mathbf{x}^5 = [30, 64, 128, 30]^\top, \quad a^5 = 1$
- $\mathbf{x}^6 = [20, 64, 128, 0]^\top, \quad a^6 = 1$

Part 1: The Sensor Grid and Heuristic Red-Teaming

TIMING		
Write-up reading, warm-up	05 min	
1.1 Sensor Grid Visualization	05 min	
1.2 Feature Space Normalization	10 min	
1.3 Competitive Event	30 min	
1.4 Scalability Analysis	10 min	
1.5 Normalization Theoretical Inquiry	10 min	

The Engineering Challenge

Before deploying statistical learning, engineering teams often attempt to write fast, manual heuristic rules. Can simple threshold logic match real physical defects, or is it fundamentally brittle?

RAW SENSOR ARRAY (2x2 PHOTODIODES)

b_{0,0}	b_{0,1}
b_{1,0}	b_{1,1}

1.1 Sensor Grid Visualization Draw the 2×2 spatial intensity layouts for all six samples in $\mathcal{D}$. Label the coordinates (r, c) and shade them according to their values (0 = black, 255 = white).

1.2 Feature Space Normalization Normalize raw pixel values to the unit floating-point interval $[0, 1]$ via min-max scaling:

$$\tilde{x}_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} = \frac{x_i}{255}$$

- Construct the normalized calibration table to three decimal places.
- Write out the augmented input vector $\tilde{\mathbf{x}}_{\text{aug}}^k \in \mathbb{R}^5$ pairing bias input $x_0 = 1$ with each sample.

1.4 Scalability Analysis In lecture notes *6b perceptron*, we examine the standard MNIST benchmark ($28 \times 28 = 784$ pixels, 60,000 images). Why does the manual heuristic approach collapse when moving from our toy 2×2 sensor to 28×28 handwritten digits? Detail three distinct mathematical and structural reasons.

1.5 Normalization Theoretical Inquiry

- Prove why running unnormalized $[0, 255]$ inputs degrades the condition number $\kappa = \frac{\lambda_{\max}}{\lambda_{\min}}$ of the design matrix covariance $\mathbf{X}^\top \mathbf{X}$ compared to normalized inputs $[0, 1]$, and state the impact on optimization.